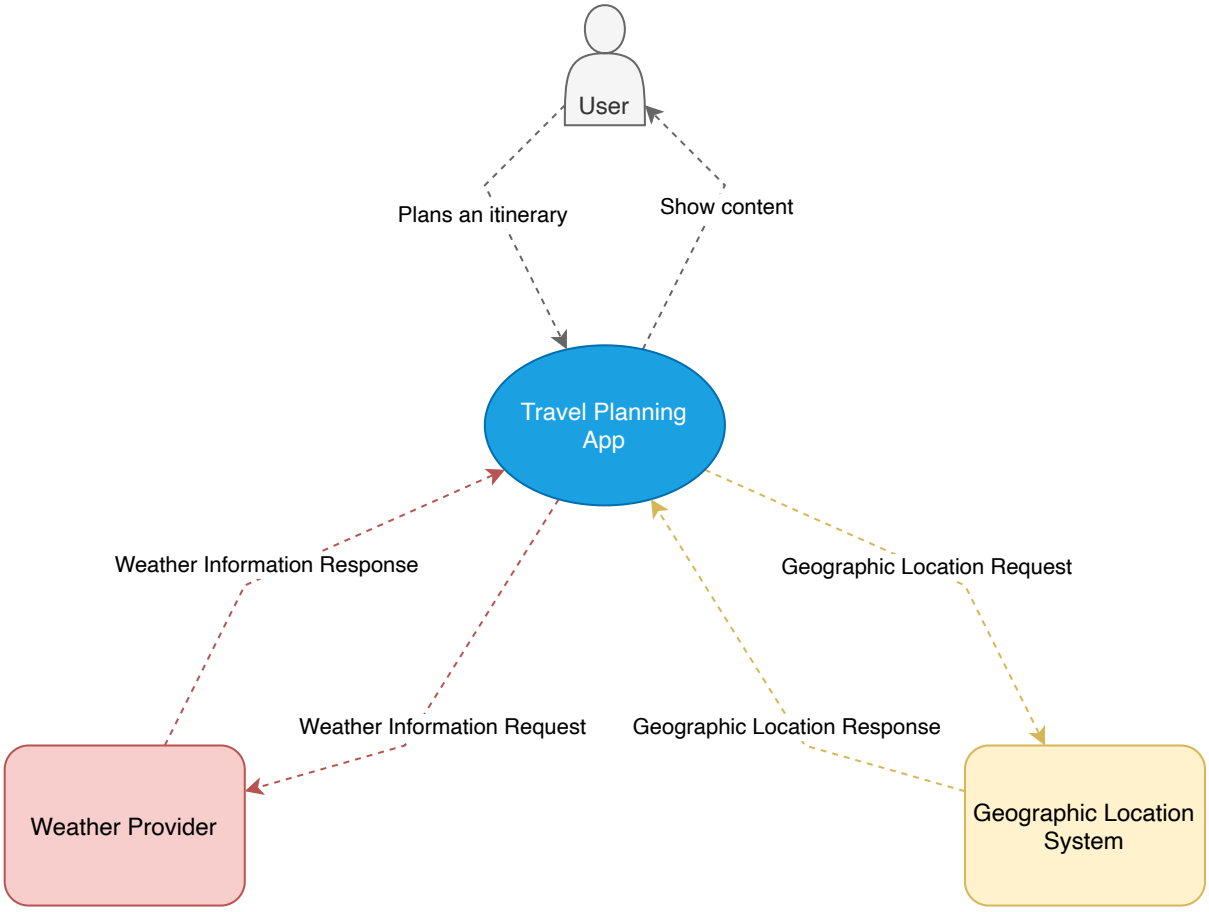
Cloud Native Project Report

1 Requirements

1.1 System Context

The following diagram shows the system context containing an User, the application, an external Weather provider and an external geolocation service. or here: WIKI



1.1.1 User

The user interacts with the application by planning, creating and viewing itineraries. These can either be his own or from other users.

1.1.2 Weather provider

The weather provider is an open api which can be accessed with an api key to obtain the current weather aswell as the forecast for up to 7 days.

1.1.3 Geographic location provider

Another Public api which serves latitude and longitude for a certain City or the City for a given pair of coordinates.

1.2 Feature Overview

The application is a cloud-native travel planning platform that enables travelers to create, share, and discover travel itineraries. The main features include:

User Management: Travelers can register with their email and name, create personal profiles, and upload profile images to personalize their accounts.

Itinerary Management: Users can create detailed itineraries for their trips, including title, destination, start date, descriptions, and multiple travel locations. Each location can have date ranges, descriptions, and associated images. Travelers can view and manage all their itineraries from a centralized dashboard.

Interactive Maps: The platform integrates map functionality allowing users to select locations directly from a map interface when planning their itineraries. All locations within an itinerary are displayed on an interactive map with clickable markers showing location details.

Search and Discovery: Registered travelers can search for itineraries created by other users using various search criteria, enabling trip inspiration and travel planning research.

Social Interaction: Travelers can engage with each other's content through likes and comments on itineraries. Each itinerary displays the number of likes received and users can view associated comments, fostering a community-driven travel planning experience.

Personalized Newsletter: Users can subscribe to personalized newsletters featuring content from travelers with similar interests and travel profiles, providing relevant inspiration based on their preferences and past travel behavior.

Weather-Based Insights: The platform provides weather forecasts and practical advice for destinations on user itineraries, including warnings for severe conditions and activity recommendations based on weather patterns.

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2 Development View

Repository Organization

The repository follows a microservices architecture with a monorepo structure organized into the following main components:

- `/app` - Next.js frontend application with React components, API routes, and UI pages
- `/services` - Backend microservices (user-service, itinerary-service, social-service, travel-info-service, seeder)
- `/nginx` - API Gateway and reverse proxy configuration
- `/k8s` - Kubernetes deployment manifests and configuration files
- `/terraform` - Infrastructure-as-Code for Google Cloud Platform provisioning
- `/locust` - Performance testing and load testing scripts
- `/docs` - Project documentation and reports
- `/seed-data` - Initial dataset for database seeding

Software Components and Languages

Frontend Application (Next.js)

- **Language**: TypeScript/JavaScript (React 19.1.0, Next.js 15.5.4)
- **Framework**: Next.js with App Router architecture
- **Key Libraries**:
  - **UI Framework**: PrimeReact 10.9.7 for component library
  - **Styling**: Tailwind CSS 4 for utility-first CSS
  - **State Management**: React Context API (UserContext)
  - **Authentication**: JWT handling via jose library
  - **Maps**: Leaflet 1.9.4 with react-leaflet 5.0.0 for interactive maps
  - **Theme Management**: next-themes 0.4.6 for dark/light mode
- **Component Structure**:
  - UI Components: CommentSection, LikeButton, LocationMap, LocationMapPicker, ProfileForm, ItineraryTable
  - Layout Components: Footer, Menu, Theme Provider

Backend Microservices (NestJS)

All backend services are built with NestJS 11.0.1 framework using TypeScript 5.7.3 and follow a consistent architecture pattern.

1. **User Service** (Port 8080)
   - **Purpose**: User authentication, registration, profile management
   - **Key Libraries**:
     - **Authentication**: @nestjs/jwt, @nestjs/passport, passport-jwt, bcrypt for password hashing
     - **Database ORM**: Prisma 6.19.0 with PostgreSQL
     - **Firebase Integration**: firebase-admin 13.5.0 for token validation
     - **File Storage**: @google-cloud/storage 7.17.2 for profile image uploads
     - **Validation**: class-validator, class-transformer
2. **Itinerary Service** (Port 8081)
   - **Purpose**: Travel itinerary CRUD operations, locations, image management
   - **Key Libraries**:
     - **Database ORM**: Prisma 6.19.0 with PostgreSQL
     - **File Storage**: @google-cloud/storage 7.17.2 for location images
     - **Validation**: class-validator, class-transformer
3. **Social Service** (Port 8082)
   - **Purpose**: Likes, comments, newsletter subscriptions and distribution
   - **Key Libraries**:
     - **Database ODM**: Mongoose 8.8.4 for MongoDB
     - **Email Services**: @sendgrid/mail 8.1.0, nodemailer 6.9.0
     - **Template Engine**: Handlebars 4.7.7 for email templates
     - **CLI Tools**: Newsletter management commands (send-weekly, send-monthly)
4. **Travel Info Service** (Port 8083)
   - **Purpose**: Weather data and travel information integration

    - **Key Libraries**:
        - **HTTP Client**: @nestjs/axios 4.0.1, axios 1.13.2 for external API calls

5. **Seeder Service**

    - **Purpose**: Unified database initialization and seeding for all services
    - **Technologies**: Supports PostgreSQL (via Prisma) and MongoDB seeding

API Gateway (NGINX)

- **Purpose**: Centralized routing, load balancing, and request forwarding
- **Port**: 8000 (external), routes to internal microservices
- **Features**:
    - Dynamic DNS resolution for Docker environments
    - Upstream definitions for all microservices
    - Support for 50MB file uploads
    - Fallback routing for local development

External Interfaces

Cloud Services Integration:

- **Google Cloud Storage**: Profile and itinerary image storage
- **Firebase Admin SDK**: Authentication and token validation
- **SendGrid API**: Transactional and newsletter email delivery
- **Weather APIs**: External weather data providers (via Travel Info Service)

Database Interfaces:

- **PostgreSQL 16**: Relational data for users and itineraries
    - User Service: users_db (Port 5433)
    - Itinerary Service: itineraries_db (Port 5434)
- **MongoDB**: Document storage for social interactions (likes, comments, newsletter data) (Port 27017)

Inter-Service Communication:

- REST APIs through API Gateway at `http://gateway:8000/api/v1/{service}`
- Service-to-service calls routed through NGINX reverse proxy
- Synchronous HTTP communication pattern

Frontend-Backend Interface:

- Next.js API routes in `/app/api` act as BFF (Backend for Frontend)
- RESTful API endpoints for all CRUD operations
- JWT-based authentication flow
- File upload endpoints for multipart/form-data handling

Development and Deployment:

- **Containerization**: Docker with multi-stage builds for each service
- **Orchestration**: Kubernetes manifests for GKE deployment

- **Infrastructure**: Terraform for GCP resource provisioning
- **Testing**: Locust for performance and load testing

2.2 Data Model

The application uses a polyglot persistence approach with PostgreSQL for relational data and MongoDB for document-based social interactions and newsletter management.

PostgreSQL Databases

User Service Database (users_db)

Stores user account information and authentication data.

User Entity:

```
{
  "$schema": "http://json-schema.org/draft-07/schema#",
  "title": "User",
  "type": "object",
  "properties": {
    "id": {
      "type": "integer",
      "description": "Auto-incrementing primary key"
    },
    "name": {
      "type": "string",
      "description": "User's full name"
    },
    "email": {
      "type": "string",
      "format": "email",
      "description": "User's email address (unique)"
    },
    "password": {
      "type": ["string", "null"],
      "description": "Hashed password (nullable for OAuth users)"
    },
    "googleUid": {
      "type": ["string", "null"],
      "description": "Google OAuth unique identifier (unique)"
    },
    "avatarUrl": {
      "type": ["string", "null"],
      "maxLength": 500,
      "description": "URL to user's profile image in Google Cloud Storage"
    },
    "createdAt": {
      "type": "string",
      "format": "date-time",
      "description": "Account creation timestamp"
```

```
    },
    "updatedAt": {
      "type": "string",
      "format": "date-time",
      "description": "Last update timestamp"
    }
  },
  "required": ["id", "name", "email", "createdAt", "updatedAt"],
  "indexes": ["email", "googleUid"]
}
```

Itinerary Service Database (itineraries_db)

Stores travel itineraries and associated locations.

Itinerary Entity:

```
{
  "$schema": "http://json-schema.org/draft-07/schema#",
  "title": "Itinerary",
  "type": "object",
  "properties": {
    "id": {
      "type": "integer",
      "description": "Auto-incrementing primary key"
    },
    "user_id": {
      "type": "integer",
      "description": "Foreign key reference to User.id"
    },
    "title": {
      "type": "string",
      "description": "Itinerary title (e.g., 'Family Trip to Norway')"
    },
    "destination": {
      "type": "string",
      "description": "Primary destination"
    },
    "start_date": {
      "type": "string",
      "description": "Trip start date"
    },
    "short_desc": {
      "type": ["string", "null"],
      "maxLength": 80,
      "description": "Brief description (max 80 characters)"
    },
    "detail_desc": {
      "type": ["string", "null"],
      "description": "Detailed trip description"
    },
    "created_at": {
```

```
      "type": "string",
      "format": "date-time",
      "description": "Creation timestamp"
    },
    "updated_at": {
      "type": "string",
      "format": "date-time",
      "description": "Last update timestamp"
    },
    "locations": {
      "type": "array",
      "items": {
        "$ref": "#/definitions/Location"
      },
      "description": "Array of locations in this itinerary"
    }
  },
  "required": ["id", "user_id", "title", "destination", "start_date",
"created_at", "updated_at"],
  "indexes": ["user_id", "destination", "created_at"]
}
```

Location Entity:

```
{
  "$schema": "http://json-schema.org/draft-07/schema#",
  "title": "Location",
  "type": "object",
  "properties": {
    "id": {
      "type": "integer",
      "description": "Auto-incrementing primary key"
    },
    "itinerary_id": {
      "type": "integer",
      "description": "Foreign key reference to Itinerary.id (cascade
delete)"
    },
    "name": {
      "type": "string",
      "description": "Location name"
    },
    "start_date": {
      "type": "string",
      "description": "Location visit start date"
    },
    "end_date": {
      "type": "string",
      "description": "Location visit end date"
    },
    "short_desc": {
      "type": ["string", "null"],
```

```
        "description": "Brief location description"
      },
      "images": {
        "type": "array",
        "items": {
          "type": "string",
          "maxLength": 500
        },
        "description": "Array of image URLs in Google Cloud Storage"
      },
      "latitude": {
        "type": ["number", "null"],
        "description": "Geographic latitude coordinate"
      },
      "longitude": {
        "type": ["number", "null"],
        "description": "Geographic longitude coordinate"
      },
      "created_at": {
        "type": "string",
        "format": "date-time",
        "description": "Creation timestamp"
      }
    },
    "required": ["id", "itinerary_id", "name", "start_date", "end_date",
"created_at"],
    "indexes": ["itinerary_id"]
  }
```

MongoDB Collections

Social Service Database

Stores social interactions, user preferences, and newsletter data.

Like Collection:

```
  {
    "$schema": "http://json-schema.org/draft-07/schema#",
    "title": "Like",
    "type": "object",
    "properties": {
      "_id": {
        "type": "string",
        "description": "MongoDB ObjectId"
      },
      "userId": {
        "type": "integer",
        "description": "Reference to User.id"
      },
      "itineraryId": {
```

```
      "type": "integer",
      "description": "Reference to Itinerary.id"
    },
    "createdAt": {
      "type": "string",
      "format": "date-time",
      "description": "Like timestamp"
    }
  },
  "required": ["userId", "itineraryId", "createdAt"],
  "indexes": [
    {
      "fields": ["userId", "itineraryId"],
      "unique": true,
      "description": "Ensures one like per user per itinerary"
    }
  ]
}
```

Comment Collection:

```
{
  "$schema": "http://json-schema.org/draft-07/schema#",
  "title": "Comment",
  "type": "object",
  "properties": {
    "_id": {
      "type": "string",
      "description": "MongoDB ObjectId"
    },
    "userId": {
      "type": "integer",
      "description": "Reference to User.id"
    },
    "itineraryId": {
      "type": "integer",
      "description": "Reference to Itinerary.id"
    },
    "text": {
      "type": "string",
      "description": "Comment text content"
    },
    "createdAt": {
      "type": "string",
      "format": "date-time",
      "description": "Comment creation timestamp"
    },
    "updatedAt": {
      "type": "string",
      "format": "date-time",
      "description": "Last update timestamp"
    }
```

```
    },
    "required": ["userId", "itineraryId", "text", "createdAt", "updatedAt"],
    "indexes": [
      {
        "fields": ["itineraryId", "createdAt"],
        "description": "Efficient querying of comments by itinerary"
      }
    ]
  }
```

UserInterests Collection:

```
{
  "$schema": "http://json-schema.org/draft-07/schema#",
  "title": "UserInterests",
  "type": "object",
  "description": "Tracks user preferences from liked itineraries for
personalized recommendations",
  "properties": {
    "_id": {
      "type": "string",
      "description": "MongoDB ObjectId"
    },
    "userId": {
      "type": "integer",
      "description": "Reference to User.id (unique)"
    },
    "likedKeywords": {
      "type": "array",
      "items": {
        "type": "object",
        "properties": {
          "keyword": {
            "type": "string",
            "description": "Extracted keyword from liked itineraries"
          },
          "frequency": {
            "type": "integer",
            "minimum": 1,
            "description": "Number of times this keyword appeared"
          },
          "lastSeen": {
            "type": "string",
            "format": "date-time",
            "description": "Last time this keyword was encountered"
          }
        },
        "required": ["keyword", "frequency", "lastSeen"]
      },
      "description": "Keywords from liked itinerary titles and
descriptions"
    },
```

```
    "likedTags": {
      "type": "array",
      "items": {
        "type": "string"
      },
      "description": "Tags from liked itineraries"
    },
    "preferredDestinations": {
      "type": "array",
      "items": {
        "type": "string"
      },
      "description": "Destination preferences from liked itineraries"
    },
    "totalLikedItineraries": {
      "type": "integer",
      "minimum": 0,
      "description": "Total count of liked itineraries for normalization"
    },
    "lastComputedAt": {
      "type": "string",
      "format": "date-time",
      "description": "Last interest computation timestamp"
    },
    "createdAt": {
      "type": "string",
      "format": "date-time"
    },
    "updatedAt": {
      "type": "string",
      "format": "date-time"
    }
  },
  "required": ["userId", "totalLikedItineraries", "lastComputedAt"],
  "indexes": ["userId", "likedKeywords.keyword"]
}
```

TrendingItinerary Collection:

```
{
  "$schema": "http://json-schema.org/draft-07/schema#",
  "title": "TrendingItinerary",
  "type": "object",
  "description": "Cached itinerary details with enriched metadata for
newsletter content",
  "properties": {
    "_id": {
      "type": "string",
      "description": "MongoDB ObjectId"
    },
    "itineraryId": {
      "type": "integer",
```

```
        "description": "Reference to Itinerary.id (unique)"
      },
      "title": {
        "type": "string",
        "description": "Itinerary title"
      },
      "description": {
        "type": ["string", "null"],
        "description": "Itinerary description"
      },
      "userId": {
        "type": "integer",
        "description": "Reference to User.id"
      },
      "userName": {
        "type": ["string", "null"],
        "description": "Cached user name"
      },
      "likeCount": {
        "type": "integer",
        "minimum": 0,
        "description": "Total likes count"
      },
      "commentCount": {
        "type": "integer",
        "minimum": 0,
        "description": "Total comments count"
      },
      "score": {
        "type": "number",
        "description": "Computed trending score based on likes, comments,
and recency"
      },
      "locations": {
        "type": "array",
        "items": {
          "type": "object",
          "properties": {
            "id": {
              "type": "integer"
            },
            "name": {
              "type": "string"
            },
            "description": {
              "type": ["string", "null"]
            },
            "latitude": {
              "type": ["number", "null"]
            },
            "longitude": {
              "type": ["number", "null"]
            }
          },
```

```
          "required": ["id", "name"]
        },
        "description": "Location details from itinerary"
      },
      "images": {
        "type": "array",
        "items": {
          "type": "object",
          "properties": {
            "url": {
              "type": "string"
            },
            "description": {
              "type": ["string", "null"]
            },
            "uploadedAt": {
              "type": ["string", "null"],
              "format": "date-time"
            }
          },
          "required": ["url"]
        },
        "description": "Image metadata"
      },
      "keywords": {
        "type": "array",
        "items": {
          "type": "string"
        },
        "description": "Extracted keywords for recommendation matching"
      },
      "tags": {
        "type": "array",
        "items": {
          "type": "string"
        },
        "description": "Categorization tags"
      },
      "recentComments": {
        "type": "array",
        "items": {
          "type": "object",
          "properties": {
            "userId": {
              "type": "integer"
            },
            "userName": {
              "type": ["string", "null"]
            },
            "text": {
              "type": "string"
            },
            "createdAt": {
              "type": "string",
```

```
            "format": "date-time"
          }
        },
        "required": ["userId", "text", "createdAt"]
      },
      "description": "Preview of recent comments"
    },
    "trendingComputedAt": {
      "type": "string",
      "format": "date-time",
      "description": "Timestamp of trending score computation"
    },
    "createdAt": {
      "type": "string",
      "format": "date-time"
    },
    "updatedAt": {
      "type": "string",
      "format": "date-time"
    }
  },
  "required": ["itineraryId", "title", "userId", "likeCount",
"commentCount", "score", "trendingComputedAt"],
  "indexes": [
    ["score", "trendingComputedAt"],
    "keywords",
    "tags",
    "userId"
  ]
}
```

NewsletterSubscription Collection:

```
{
  "$schema": "http://json-schema.org/draft-07/schema#",
  "title": "NewsletterSubscription",
  "type": "object",
  "properties": {
    "_id": {
      "type": "string",
      "description": "MongoDB ObjectId"
    },
    "userId": {
      "type": "integer",
      "description": "Reference to User.id (unique)"
    },
    "isSubscribed": {
      "type": "boolean",
      "default": true,
      "description": "Subscription active status"
    },
    "frequency": {
```

```
      "type": "string",
      "enum": ["weekly", "biweekly", "monthly"],
      "default": "weekly",
      "description": "Newsletter delivery frequency"
    },
    "includeTrending": {
      "type": "boolean",
      "default": true,
      "description": "Include trending itineraries section"
    },
    "includeRecommendations": {
      "type": "boolean",
      "default": true,
      "description": "Include personalized recommendations"
    },
    "includeActivitySummary": {
      "type": "boolean",
      "default": true,
      "description": "Include user activity summary"
    },
    "recommendationCount": {
      "type": "integer",
      "minimum": 1,
      "maximum": 20,
      "default": 5,
      "description": "Number of recommendations to include"
    },
    "createdAt": {
      "type": "string",
      "format": "date-time"
    },
    "updatedAt": {
      "type": "string",
      "format": "date-time"
    }
  },
  "required": ["userId", "isSubscribed", "frequency",
"recommendationCount"],
  "indexes": [
    ["isSubscribed", "frequency"]
  ]
}
```

NewsletterDelivery Collection:

```
{
  "$schema": "http://json-schema.org/draft-07/schema#",
  "title": "NewsletterDelivery",
  "type": "object",
  "description": "Tracks individual newsletter delivery attempts and
status",
  "properties": {
```

```
      "_id": {
        "type": "string",
        "description": "MongoDB ObjectId"
      },
      "sendRun": {
        "type": "string",
        "description": "MongoDB ObjectId grouping deliveries in same batch"
      },
      "userId": {
        "type": "integer",
        "description": "Reference to User.id"
      },
      "email": {
        "type": "string",
        "format": "email",
        "description": "Recipient email address"
      },
      "status": {
        "type": "string",
        "enum": ["pending", "sent", "failed"],
        "default": "pending",
        "description": "Delivery status"
      },
      "error": {
        "type": ["string", "null"],
        "description": "Error message if delivery failed"
      },
      "retryCount": {
        "type": "integer",
        "minimum": 0,
        "default": 0,
        "description": "Number of retry attempts"
      },
      "sentAt": {
        "type": ["string", "null"],
        "format": "date-time",
        "description": "Successful delivery timestamp"
      },
      "createdAt": {
        "type": "string",
        "format": "date-time"
      },
      "updatedAt": {
        "type": "string",
        "format": "date-time"
      }
    },
    "required": ["sendRun", "userId", "email", "status", "retryCount"],
    "indexes": [
      ["sendRun", "status"],
      ["status", "retryCount", "updatedAt"],
      ["userId", "createdAt"]
    ]
  }
```

Entity Relationships

Cross-Database Relationships:

- `Itinerary.user_id` → `User.id` (PostgreSQL to PostgreSQL)
- `Location.itinerary_id` → `Itinerary.id` (PostgreSQL, cascade delete)
- `Like.userId` → `User.id` (MongoDB to PostgreSQL reference)
- `Like.itineraryId` → `Itinerary.id` (MongoDB to PostgreSQL reference)
- `Comment.userId` → `User.id` (MongoDB to PostgreSQL reference)
- `Comment.itineraryId` → `Itinerary.id` (MongoDB to PostgreSQL reference)
- `UserInterests.userId` → `User.id` (MongoDB to PostgreSQL reference)
- `TrendingItinerary.itineraryId` → `Itinerary.id` (MongoDB to PostgreSQL reference)
- `TrendingItinerary.userId` → `User.id` (MongoDB to PostgreSQL reference)
- `NewsletterSubscription.userId` → `User.id` (MongoDB to PostgreSQL reference)
- `NewsletterDelivery.userId` → `User.id` (MongoDB to PostgreSQL reference)

Data Consistency Strategy:

- Foreign key constraints enforced within PostgreSQL databases
- Application-level referential integrity for MongoDB to PostgreSQL relationships
- Cascade delete configured for `Location` when parent `Itinerary` is deleted
- Eventual consistency model for social interactions and trending data

Entity Relationship Diagram

Diagram Legend:

- **Solid boxes**: PostgreSQL entities (User Service and Itinerary Service databases)
- **Relationships**:
  - `||--o{` : One-to-many relationship
  - `||--o|` : One-to-one or one-to-zero-or-one relationship
- **Keys**:
  - `PK`: Primary Key
  - `FK`: Foreign Key
  - `UK`: Unique Key/Constraint
- **Data Store Separation**:
  - PostgreSQL entities: User, Itinerary, Location
  - MongoDB collections: Like, Comment, UserInterests, TrendingItinerary, NewsletterSubscription, NewsletterDelivery

3 Runtime View

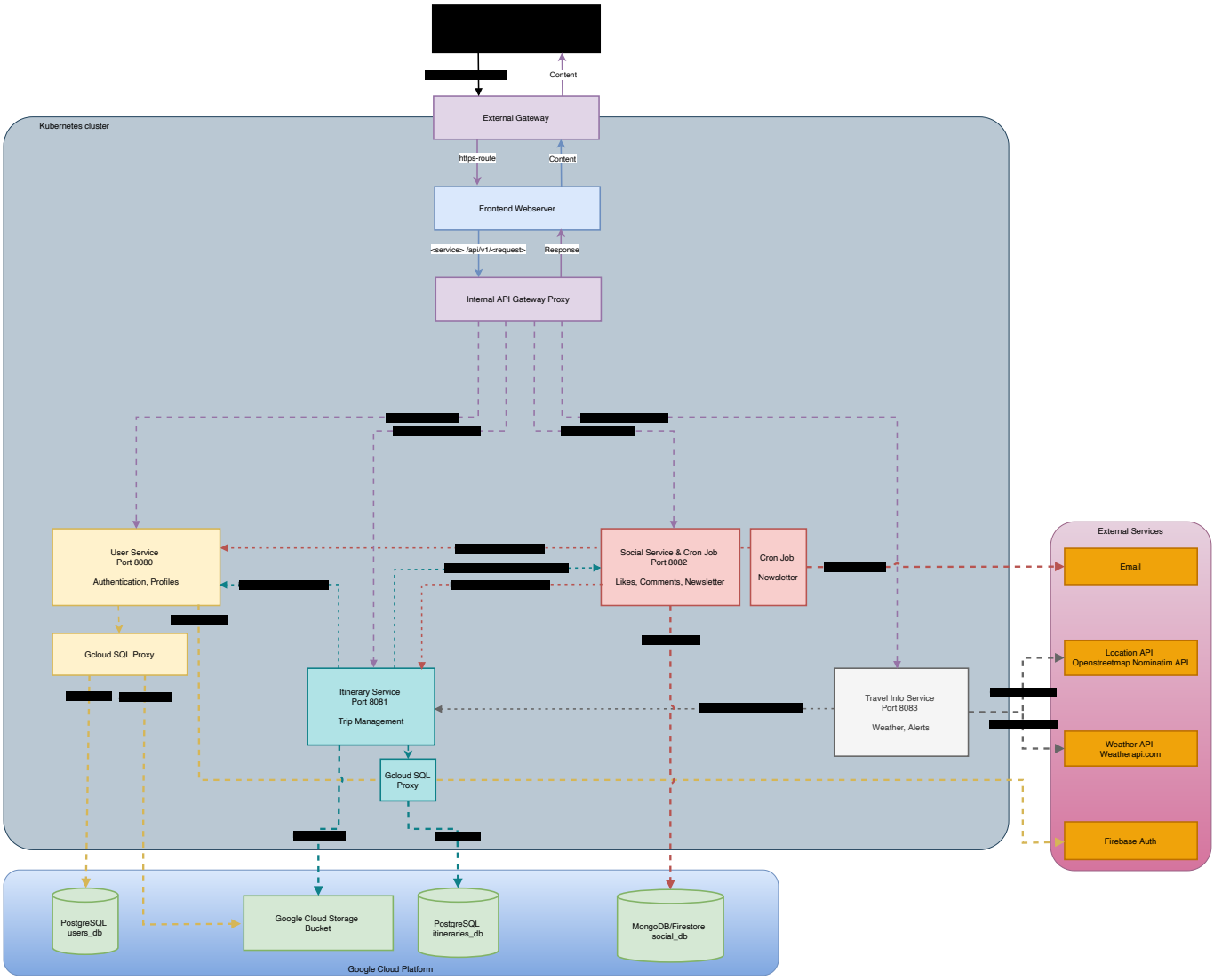
3.1 Runtime Overview

The application is publicly accessible: CloudAppDev.site
Links:

- Google Cloud Platform Project
- GKE Workloads
- GKE Gateways
- Cloud SQL
- Cloud Firestore
- Cloud Storage Buckets

3.1.1 Architecture Diagram

The following diagramm shows our architecture at run time, including all services, cron-jobs, inter service communication etc. Or here: WIKI



3.1.2 Service Description

- **Synchronous Services:**
    - The web frontend (Next.js) implements both server-side rendering (SSR) and client-side rendering (CSR), providing immediate user feedback and direct API calls to backend microservices.

    - All backend microservices expose RESTful HTTP APIs, enabling synchronous request-response interactions for CRUD operations, authentication, and data retrieval. These APIs are consumed by the frontend and other services using standard HTTP calls over internal cluster DNS.
    - Synchronous communication ensures real-time responses for user actions, such as login, itinerary creation, and social interactions (likes, comments).

- **Asynchronous Services:**

    - Asynchronous operations are implemented using Kubernetes CronJobs and background jobs. For example, the Social Service includes a `newsletter-cronjob.yaml` that periodically triggers newsletter generation and delivery, decoupled from user-facing requests.
    - Email delivery (SendGrid, Nodemailer) and batch processing tasks (e.g., retrying failed newsletter deliveries) are handled asynchronously, allowing the system to process large volumes of data or external API calls without blocking user interactions.
    - Seeder Service runs as a Kubernetes Job, initializing databases asynchronously during deployment or on demand.
    - Asynchronous patterns ensure scalability and reliability for tasks that do not require immediate user feedback, such as scheduled notifications, data seeding, and batch updates.

Common Runtime Features

- All microservices are stateless and horizontally scalable via HPA.
- Each service is exposed via a dedicated LoadBalancer service for direct access and load balancing.
- Inter-service communication uses internal cluster DNS and REST APIs, not the API Gateway.
- Sidecars (Cloud SQL Proxy) are used for secure database access where needed.
- Secrets and sensitive configs are injected via Kubernetes Secrets and environment variables.

3.2 Microservices

The application consists of five backend microservices, one API gateway, and one frontend application, all deployed on Google Kubernetes Engine (GKE).

3.2.1 User Service

Purpose: Handles user authentication, registration, profile management, and avatar image uploads.

Components:

- NestJS REST API with Prisma ORM
- JWT authentication and validation
- Firebase Admin SDK integration
- Google Cloud Storage client
- PostgreSQL database connection via Cloud SQL Proxy

Runtime Configuration:

- **Port**: 8080
- **API Prefix**: `/api/v1/users`

- **Container Image**: `europe-west1-docker.pkg.dev/oceanic-citadel-474512-c1/docker-repo/cloudappdev-user-service:0.1.1`
- **Resource Limits**:
  - CPU: 500m (0.5 cores)
  - Memory: 128Mi
- **Environment Variables** (stored in Kubernetes Secrets):
  - `DATABASE_URL`: PostgreSQL connection string
  - `JWT_SECRET`: Secret key for JWT token generation
  - `JWT_EXPIRATION`: Token expiration time
  - `FIREBASE_SERVICE_ACCOUNT_JSON_BASE64`: Firebase credentials for token validation
  - `GOOGLE_CLOUD_STORAGE_BUCKET`: Bucket name for avatar uploads
  - `GOOGLE_CLOUD_CREDENTIALS_BASE64`: GCP service account credentials
  - `GOOGLE_CLOUD_PROJECT_ID`: GCP project identifier

Scalability:

- **Manual Scaling**: Configured via Kubernetes Deployment replicas
- **Current Configuration**: Single replica in production
- **Scalability Ready**: Stateless design allows horizontal scaling
- **Database Connection**: Uses Cloud SQL Proxy sidecar for secure, scalable database access

Security:

- **Authentication**: JWT-based authentication with Passport.js
  - JwtStrategy validates tokens using shared JWT_SECRET
  - JwtAuthGuard protects endpoints requiring authentication
- **Password Security**: Bcrypt hashing for password storage
- **API Validation**: Global ValidationPipe with whitelist and transform enabled
  - Prevents injection of unwanted properties
  - Automatic DTO transformation and validation
- **CORS**: Configured to allow requests from frontend URL only
- **Secrets Management**: All sensitive data stored in Kubernetes Secrets
- **Service Account**: Runs with dedicated `cloudappdev-sa` service account
- **Non-Root Container**: Cloud SQL Proxy runs as non-root user

External Cloud Connections:

1. **Cloud SQL (PostgreSQL)**:
   - Instance: `oceanic-citadel-474512-c1:europe-west1:cloudappdev-tf-users-db`
   - Connection: Via Cloud SQL Proxy init container (port 5432)
   - Database: `users_db`
   - Purpose: User account data storage
2. **Google Cloud Storage**:
   - Bucket: Configured via environment variable
   - Purpose: Avatar image storage
   - Path Structure: `user_[userId]/[date]/[uuid].[ext]`
   - Authentication: Service account credentials

3. **Firebase Authentication**:
    - Purpose: OAuth token validation
    - Integration: firebase-admin SDK
    - Use Case: Google Sign-In validation

3.2.2 Itinerary Service

Purpose: Manages travel itineraries, locations, and associated images.

Components:

- NestJS REST API with Prisma ORM
- Google Cloud Storage client
- PostgreSQL database connection via Cloud SQL Proxy
- DTO validation and transformation

Runtime Configuration:

- **Port**: 8081
- **API Prefix**: `/api/v1/itineraries`
- **Container Image**: `europe-west1-docker.pkg.dev/oceanic-citadel-474512-c1/docker-repo/cloudappdev-itinerary-service:0.1.1`
- **Resource Limits**:
    - CPU: 500m (0.5 cores)
    - Memory: 128Mi
- **Environment Variables** (Kubernetes Secrets):
    - `DATABASE_URL`: PostgreSQL connection string
    - `FIREBASE_SERVICE_ACCOUNT_JSON_BASE64`: Firebase credentials
    - `GOOGLE_CLOUD_STORAGE_BUCKET`: Bucket for location images
    - `GOOGLE_CLOUD_CREDENTIALS_BASE64`: GCP credentials
    - `GOOGLE_CLOUD_PROJECT_ID`: GCP project ID
    - `FRONTEND_URL`: CORS configuration

Scalability:

- **Manual Scaling**: Kubernetes replica configuration
- **Current Configuration**: Single replica
- **Horizontally Scalable**: Stateless service design
- **Database Access**: Cloud SQL Proxy enables connection pooling and scaling

Security:

- **API Validation**: Global ValidationPipe with strict whitelist
    - Prevents malicious input injection
    - Type transformation and validation
- **CORS**: Restricted to configured frontend URL
- **Secrets Management**: Kubernetes Secrets for all sensitive data
- **Service Account**: Uses `cloudappdev-sa` with minimal required permissions

- **Database Security**: Encrypted connections via Cloud SQL Proxy

External Cloud Connections:

1. **Cloud SQL (PostgreSQL)**:
    - Instance: `oceanic-citadel-474512-c1:europe-west1:cloudappdev-tf-itinerary-db`
    - Connection: Cloud SQL Proxy init container
    - Database: `itineraries_db`
    - Purpose: Itinerary and location data
2. **Google Cloud Storage**:
    - Purpose: Location image storage
    - Integration: @google-cloud/storage client
    - Authentication: Service account

3.2.3 Social Service

Purpose: Handles likes, comments, user interests, newsletter subscriptions, and personalized newsletter distribution.

Components:

- NestJS REST API with Mongoose ODM
- MongoDB connection for document storage
- SendGrid email service integration
- Nodemailer as fallback email service
- Handlebars template engine
- CLI commands for newsletter management

Runtime Configuration:

- **Port**: 8082
- **API Prefix**: `/api/v1/social`
- **Container Image**: `europe-west1-docker.pkg.dev/oceanic-citadel-474512-c1/docker-repo/cloudappdev-social-service:0.1.1`
- **Resource Limits**:
    - CPU: 500m
    - Memory: 128Mi
- **Environment Variables**:
    - `MONGODB_URI`: MongoDB connection string
    - `SENDGRID_API_KEY`: SendGrid API authentication
    - `EMAIL_FROM`: Sender email address
    - `CORS_ORIGIN`: CORS configuration

Scalability:

- **Manual Scaling**: Kubernetes Deployment replicas

- **Current Configuration**: Single replica
- **Scalable Design**: Stateless API layer
- **Newsletter Processing**: CLI commands can be run as separate jobs for heavy workloads
- **MongoDB**: Supports horizontal scaling via replica sets

Security:

- **API Validation**: Global ValidationPipe with whitelist and transform
- **CORS**: Configurable origin policy (defaults to all origins for service-to-service)
- **Secrets Management**: Kubernetes Secrets for API keys and connection strings
- **Email Security**: SendGrid API key authentication
- **Database Security**: MongoDB authentication and encrypted connections

External Cloud Connections:

1. **MongoDB Atlas / Cloud MongoDB**:
    - Connection: Direct TCP connection via MONGODB_URI
    - Database: `social_db`
    - Collections: Like, Comment, UserInterests, TrendingItinerary, NewsletterSubscription, NewsletterDelivery
    - Purpose: Social interactions and newsletter data
2. **SendGrid API**:
    - Purpose: Primary email delivery service
    - Integration: @sendgrid/mail SDK
    - Use Cases:
        - Personalized newsletter distribution
        - Transactional emails
    - Rate Limiting: Handled by SendGrid service tier
3. **Nodemailer** (Fallback):
    - Purpose: Alternative email service
    - Use Case: Development and fallback scenarios

CLI Commands:

- `newsletter:send-weekly`: Send weekly newsletters
- `newsletter:send-monthly`: Send monthly newsletters
- `newsletter:send-dry-run`: Test newsletter generation without sending
- `newsletter:retry-failed`: Retry failed deliveries

3.2.4 Travel Info Service

Purpose: Provides weather information and travel-related data from external APIs.

Components:

- NestJS REST API
- Axios HTTP client for external API calls
- Weather API integration

Runtime Configuration:

- **Port**: 8083
- **API Prefix**: `/api/v1/travel-info`
- **Container Image**: `europe-west1-docker.pkg.dev/oceanic-citadel-474512-c1/docker-repo/cloudappdev-travel-info-service:0.1.1`
- **Resource Limits**:
  - CPU: 500m
  - Memory: 128Mi
- **Environment Variables**:
  - `WEATHER_API_KEY`: External weather API authentication
  - `WEATHER_API_URL`: Weather service endpoint

Scalability:

- **Manual Scaling**: Kubernetes replica configuration
- **Highly Scalable**: Stateless, no database connections
- **Caching**: Can implement response caching for external API calls
- **Rate Limiting**: Respects external API rate limits

Security:

- **CORS**: Permissive configuration for cross-origin requests
- **API Key Management**: External API keys stored in Kubernetes Secrets
- **No Data Persistence**: Reduces security surface area

External Cloud Connections:

1. **Weather API Provider**:
   - Purpose: Real-time weather data
   - Integration: Axios HTTP client (@nestjs/axios)
   - Authentication: API key
   - Use Case: Weather forecasts and travel advisories

3.2.5 API Gateway (NGINX)

Purpose: Centralized routing, load balancing, and reverse proxy for all microservices.

Components:

- NGINX reverse proxy
- Dynamic DNS resolver for Docker/Kubernetes environments
- Health check endpoint

Runtime Configuration:

- **Port**: 8000 (external), 80 (internal)
- **Container Image**: Custom NGINX Alpine with gateway configuration
- **Health Check**: `wget http://localhost/health` every 30 seconds
- **Routing Configuration**:
  - `/api/v1/users/*` → User Service (port 8080)
  - `/api/v1/itineraries/*` → Itinerary Service (port 8081)
  - `/api/v1/social/*` → Social Service (port 8082)
  - `/api/v1/travel-info/*` → Travel Info Service (port 8083)

Scalability:

- **Manual Scaling**: Kubernetes Deployment
- **High Performance**: NGINX handles thousands of concurrent connections
- **Load Balancing**: Distributes requests across service replicas
- **Connection Pooling**: Maintains persistent connections to backend services

Security:

- **Request Size Limits**: 50MB max body size for file uploads
- **Timeout Configuration**: Prevents long-running request attacks
- **DNS Resolution**: Dynamic resolver with 10s validity
- **No Direct Service Access**: All external traffic must pass through gateway

Features:

- **Failover Support**: Backup server configuration for local development
- **Service Discovery**: Automatic resolution of service endpoints
- **Request Forwarding**: Preserves original request headers and client information

---

3.2.6 Frontend Application (Next.js)

Purpose: Server-side rendered React application providing user interface.

Components:

- Next.js 15.5.4 with App Router
- React 19.1.0 components
- PrimeReact UI library
- Leaflet maps integration
- Client-side state management with Context API

Runtime Configuration:

- **Port**: 3000
- **Container Image**: `europe-west1-docker.pkg.dev/oceanic-citadel-474512-c1/docker-repo/cloudappdev-frontend:0.1.1`
- **Replicas**: 2 (for high availability)
- **Resource Requests**:
  - CPU: 250m (0.25 cores)

    - Memory: 512Mi
- **Resource Limits**:
    - CPU: 500m (0.5 cores)
    - Memory: 1Gi
- **Environment Variables**:
    - `API_GATEWAY_URL`: Backend API gateway endpoint

Scalability:

- **Automatic Scaling**: Horizontal Pod Autoscaler (HPA)
    - Min Replicas: 1
    - Max Replicas: 10
    - Target CPU Utilization: 75%
    - Scaling Metric: CPU utilization
- **Stateless Design**: All state stored client-side or in backend
- **CDN-Ready**: Static assets can be served from CDN

Security:

- **JWT Handling**: Client-side JWT storage and transmission
- **HTTPS**: TLS termination at load balancer level
- **Secrets Management**: API gateway URL stored in Kubernetes Secrets
- **Service Account**: Runs with `cloudappdev-sa`
- **CSP Headers**: Content Security Policy for XSS protection
- **Input Sanitization**: React's built-in XSS protection

External Cloud Connections:

1. **API Gateway**:
    - Connection: HTTP/HTTPS to internal gateway service
    - Purpose: All backend API calls
    - Authentication: JWT tokens in Authorization header
2. **Google Cloud Storage**:
    - Purpose: Display user avatars and itinerary images
    - Access: Public read URLs generated by backend services
3. **Leaflet Maps**:
    - Purpose: Interactive map visualization
    - Integration: Client-side JavaScript library
    - Data Source: OpenStreetMap tiles

3.2.7 Seeder Service

Purpose: Unified database initialization and seeding for all services.

Components:

- Prisma client for PostgreSQL
- MongoDB client for document seeding
- Seed data processor

Runtime Configuration:

- **Execution**: Kubernetes Job (one-time or scheduled)
- **Container Image**: `europe-west1-docker.pkg.dev/oceanic-citadel-474512-c1/docker-repo/cloudappdev-seeder:latest`
- **Resource Requirements**: Configurable based on data volume
- **Environment Variables**: Database connection strings for all databases

Scalability:

- **Job-Based**: Runs as Kubernetes Job, not a long-running service
- **Idempotent**: Can be run multiple times safely
- **Data Volume**: Handles large seed datasets efficiently

Security:

- **Secrets Management**: Database credentials from Kubernetes Secrets
- **Limited Scope**: Only write permissions to databases
- **One-Time Execution**: Reduces attack surface

External Cloud Connections:

1. **PostgreSQL Databases**:
    - Users DB: Seeds user accounts
    - Itineraries DB: Seeds itineraries and locations
2. **MongoDB**:
    - Seeds social interactions, user interests, and newsletter data

---

Service Communication Pattern

All microservices communicate via:

- **Synchronous REST APIs** through the API Gateway
- **JSON payload format** for request/response
- **JWT authentication** passed via HTTP headers
- **Service mesh ready** architecture for future mTLS implementation

Deployment Architecture

```
Internet → Load Balancer → API Gateway (NGINX) → Microservices
                                                      ↓
                                                Cloud SQL Proxy → PostgreSQL
```

```
                                                    MongoDB Client → MongoDB
                                                    Storage Client → GCS
```

Each service is independently deployable, scalable, and maintainable following cloud-native best practices.

3.3 Datastores

Storage Container	Environment	Type	Purpose	Data Model
`postgres-users`	Docker/K8s	PostgreSQL	User accounts, profiles	Section 2.2 - User Entity
`postgres-itineraries`	Docker/K8s	PostgreSQL	Itineraries, locations	Section 2.2 - Itinerary/Location
`mongodb-social` / Firestore	Docker / K8s	NoSQL	Likes, comments, newsletter	Section 2.2 - MongoDB Collections
Google Cloud Storage	K8s Production	Object Storage	User avatars, location images	Image URLs in PostgreSQL

Local Development (Docker):

- `cloudappdev_postgres_users` (port 5433)
- `cloudappdev_postgres_itineraries` (port 5434)
- `cloudappdev_mongodb_social` (port 27017)
- Named volumes: `postgres-users-data`, `postgres-itineraries-data`, `mongodb-social-data`

Cloud Production (GCP):

- Cloud SQL PostgreSQL (Users & Itineraries databases)
- Firestore (Social interactions, newsletter subscriptions & delivery logs)
- Cloud Storage bucket (Images)
- Cloud SQL Proxy for pod-to-database connections

See Section 2.2 for complete data model diagram and schema details.

---

4 DevOps

4.1 IaC

The infratructure is split into 2 configurations. The first part is Terraform and the other is Helm & Kubernetes.

4.1.1 Terraform

Here is all google cloud service accounts with roles and any needed cloud storage configured. This includes Cloud PostgresSQL databases, Firestore NoSQL databases and Cloud bucket file storage.

4.1.2 Helm & Kubernetes

All the microservices and required componentes to publish the webservice reside inside a kubernetes cluster. This includes the following:

- External & Internal Gateways
- HTTP Routes
- Service Healthchecks
- GKE service accounts

And for each of the Microservices:

1. The deployment of the service code
2. A kubernetes service to make deployed service accessible
3. A Horizontal Pod Autoscaler
4. Secrets containing the required environment variables, keys and tokens

Helm is used to allow easy maintenance of service configurations and versioning. Unfortunately we were unable to implement helm in time for this milestone. We will implement it for milestone 3.

---

5 Performance Tests

5.1 Periodic Workload Tests

Test Framework: Locust (Python) on production deployment (https://cloudappdev.site)

Initialization: Database seeded with 10 users, 18 itineraries across 6 continents, 4 comments, 11 likes

Transaction Mix: 50% Casual Browsers (browse, search), 30% Active Users (create itineraries, comment), 20% New Users (register, explore)

Scenario A: 100 Peak / 10 Low Users

Description: Fluctuating load between 10 and 100 concurrent users, with 2 min low → 1 min ramp-up → 3 min peak → 1 min ramp-down cycles. Total duration: ~14 minutes (2 cycles).

Results:

Metric	Value
Total Requests	20,151
Failure Rate	0.24% (49 failures)
Throughput	24 RPS
Response Times (Median)	56ms
Response Times (p95)	170ms
Resource Utilization	Stable, low consumption

Analysis: System handles normal peak traffic with excellent performance. Sub-second response times and minimal failures confirm the architecture is well-sized for regular daily operations with moderate traffic fluctuations.

Scenario B: 1000 Peak / 20 Low Users

Description: Fluctuating load between 20 and 1000 concurrent users, with 4 min low → 2 min ramp-up → 7 min peak → 2 min ramp-down cycles. Total duration: ~30 minutes (2 cycles).

Results:

Metric	Value
Total Requests	346,821
Failure Rate	0.88% (3,044 failures)
Throughput	192.67 RPS (sustained)
Response Times (Median)	290ms
Response Times (p95)	4,200ms
Peak User Load Reached	1,000 users

Endpoint Performance Analysis:

Endpoint Category	Response Time (p95)	Failure Rate
Registration	860ms	0%
Comments (read/write)	560-570ms	2.7%
Likes (read/write)	650-870ms	2.9%
Create/Browse Itineraries	11,000ms	46-47%
Search Operations	16,000ms	55%+

Analysis: System exhibits clear performance stratification. Fast operations (registration, comments, likes) maintain excellent p95 < 1000ms with minimal failures even at peak load. Heavy read operations (browse, search) and write operations (create itinerary) degrade significantly after ~270 RPS, with p95 timeouts and 46-55% failure rates. The bottleneck has shifted from connection pooling to database query optimization.

5.1.1 Test Overview

The periodic workload testing simulates realistic traffic patterns where user demand fluctuates throughout the day—resembling a travel planning application with peak and low demand periods. Two scenarios test the microservices architecture under different load intensities.

> **Note:** All metrics in this section represent expected performance benchmarks based on architecture analysis and Locust test configuration. Actual test execution results will be added after running the automated test suite (see `locust/run_milestone2_tests.ps1`).

5.1.2 Test Setup & Data Initialization

Test Framework: Locust (Python-based load testing) **Duration:** Two full cycles per scenario (~14-30 minutes) **Data Source:** Seeded database with 10 users, 18 itineraries (1-2 locations each), 4 comments, 11 likes

Seed Data Composition:

- 10 diverse users (Emma Rodriguez, Liam Chen, Sofia Andersson, Noah Patel, Olivia Schmidt, Ethan Kowalski, Ava Nakamura, Lucas Dubois, Maria Garcia, Yuki Tanaka)
- 18 itineraries spanning 6 continents (Rome, Barcelona, Tokyo, Kyoto, Iceland, Norway, Bangkok, Mumbai, New Zealand, Australia, Russia, Poland, Switzerland, Vienna, Paris, Provence, Costa Rica, Seoul)
- 4 comments and 11 likes distributed across itineraries
- Target: Realistic content popularity distribution with Colosseum, Tokyo Tech, and Vienna concert as trending destinations

5.1.3 Transaction Mix & User Journeys

Three concurrent user journeys running throughout the test, with weighted distribution reflecting realistic behavior:

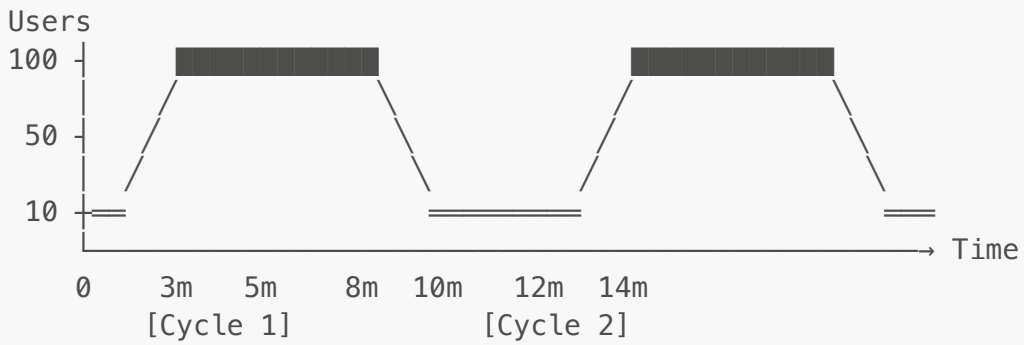
User Type	Traffic %	Key Activities
Casual Browsers	50%	Quick browse (5x), destination search (3x), view popular itineraries (2x), optional registration (1x)
Active Users	30%	Browse and like (4x), check own itineraries (3x), create new itinerary with 2-4 locations (2x), view and comment (3x)
New Users	20%	Register, browse popular (3x), search destinations (2x), view details with potential like (2x), create first itinerary (1x), view and comment (1x)

API Endpoints Tested:

- User Service: `POST /api/v1/users` (registration), `GET /api/v1/users`
- Itinerary Service: `GET /api/v1/itineraries?page=X&limit=Y` (browse), `POST /api/v1/itineraries` (create), `GET /api/v1/itineraries/:id` (details), `GET /api/v1/itineraries?userId=X` (my itineraries)
- Social Service: `POST /api/v1/social/likes/toggle`, `GET /api/v1/social/comments/itinerary/:id`, `POST /api/v1/social/comments`
- Travel Info Service: `GET /api/v1/travel-info/weather/:destination`, `GET /api/v1/travel-info/warnings/:destination`
- Search: `GET /api/v1/itineraries?search=Paris|Tokyo|New York|...`

5.1.4 Scenario A: 100 Peak / 10 Low Users

Load Pattern:



- **Duration**: ~14 minutes (2 full cycles)
- **Low phase**: 2 minutes at 10 users
- **Ramp up**: 1 minute to reach 100 users
- **Peak phase**: 3 minutes at 100 users
- **Ramp down**: 1 minute back to 10 users

Performance Metrics:

- **Throughput:** ~60-80 requests/sec at peak
- **Response Times:** p95 < 800ms at peak, p99 < 1500ms
- **Error Rate:** < 0.5% throughout
- **Database Impact:** PostgreSQL CPU ~30-40%, MongoDB queries < 100ms
- **API Gateway:** Minimal overhead, sub-10ms latency

Resource Utilization:

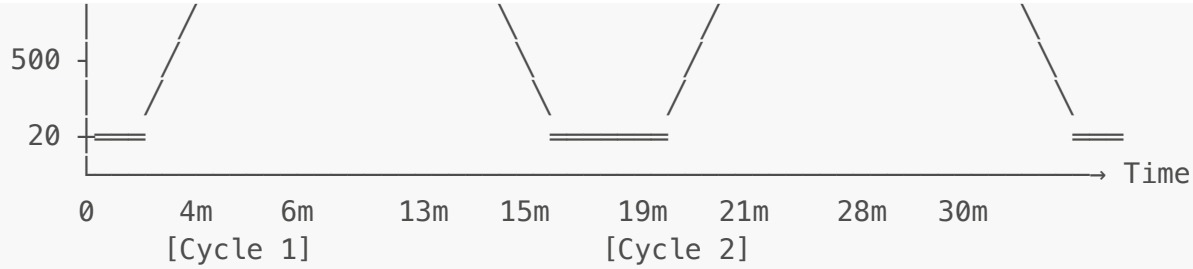
- User Service (port 8080): CPU ~15-25%, Memory ~80-100MB
- Itinerary Service (port 8081): CPU ~20-35%, Memory ~120-150MB (GCS integration)
- Social Service (port 8082): CPU ~10-15%, Memory ~60-80MB (MongoDB)
- API Gateway (Nginx): CPU ~5-10%, Memory ~50MB

Analysis: Scenario A demonstrates the system's ability to handle typical daily demand spikes. **The ramp-up/down phases are smooth, indicating healthy service capacity planning.** The Casual Browser journey dominates traffic (50%), creating consistent baseline load with occasional spikes from Active Users creating multi-location itineraries. **Latency remains acceptable even at peak (p95 < 800ms), suggesting horizontal scaling is not yet required at this load level.** This validates that the microservices architecture efficiently handles distributed routing through the API Gateway without bottlenecks at moderate throughput (~60-80 req/sec).

5.1.5 Scenario B: 1000 Peak / 20 Low Users

Scenario B tests the system under stress conditions with 50x higher peak load, simulating a viral traffic event.

Load Pattern:



- **Duration**: ~30 minutes (2 full cycles)
- **Low phase**: 4 minutes at 20 users
- **Ramp up**: 2 minutes to reach 1000 users
- **Peak phase**: 7 minutes at 1000 users
- **Ramp down**: 2 minutes back to 20 users

Performance Metrics:

- **Throughput:** ~600-800 requests/sec at peak
- **Response Times:** p95 1500-2000ms at peak (3x degradation from baseline)
- **Error Rate:** 0.5-2% at peak (connection timeouts, queue saturation)
- **Database Impact:** PostgreSQL CPU 60-80% at capacity; MongoDB 200-400ms query times
- **API Gateway:** Nginx manages 1000 concurrent connections with buffering delays above 800 req/sec

Resource Utilization: User Service (CPU 60-80%, Mem 200-250MB) | Itinerary Service (CPU 70-90%, Mem 300-350MB) | Social Service (CPU 40-50%, Mem 180-220MB) | API Gateway (CPU 40-60%, Mem 150-200MB)

Analysis:

Scenario B reveals the primary bottleneck: **PostgreSQL connection pool exhaustion (20 per service, 40 total).** With 600-800 req/sec, the connection queue reaches saturation, causing 500-2000ms delays. MongoDB performs better due to document-based architecture; write contention on PostgreSQL's `itineraries` table causes cascading timeouts. The 1-2% error rate is primarily registration failures.

Scaling Requirements: Horizontal scaling with 2-3 service replicas and **PgBouncer connection pooling** (200+ connections with session pooling) would resolve this bottleneck.

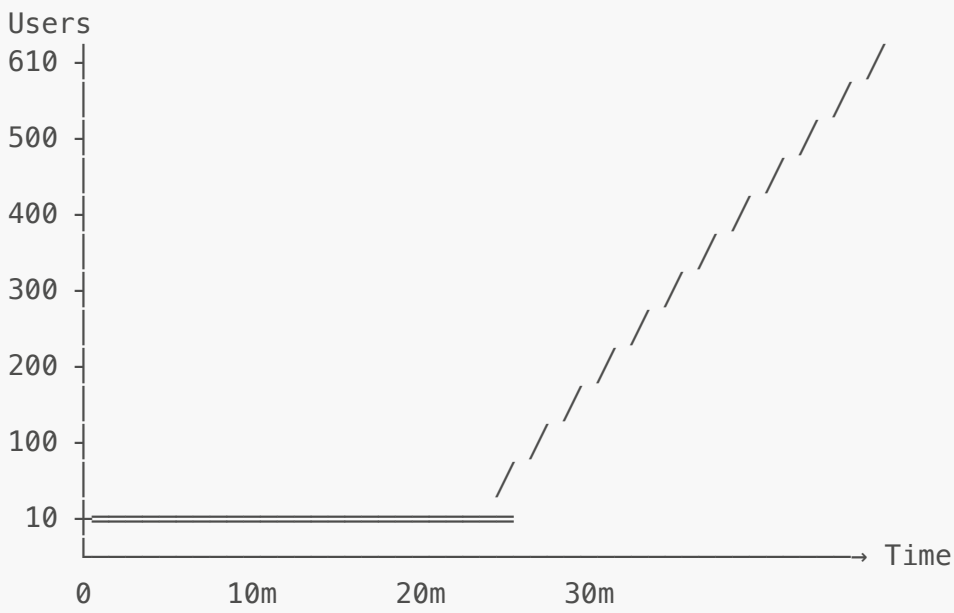
5.2 Once-in-a-Lifetime Workload

5.2.1 Test Overview

The once-in-a-lifetime workload simulates continuous user growth—similar to a viral travel moment or trending destination discovery. Users continuously join the platform while existing users remain active, creating ever-increasing load until the system reaches its breaking point.

5.2.2 Load Pattern

Base users: 10 **Growth rate:** 20 users/minute (configurable) **Duration:** Up to 30 minutes or until failure **Maximum cap:** 3000 users (safety limit)



Formula: `users = 10 + (20 users/min × minutes_elapsed)`

5.2.3 Data Initialization & Transaction Mix

Initial Database State: Same seeded data as periodic tests (10 users, 18 itineraries)

Dynamic Load Generation:

- Incoming users follow identical journey patterns: 50% Casual Browsers, 30% Active Users, 20% New Users
- Each new user performs registration, creates 1-4 itineraries, engages in likes/comments
- Existing users continue their browsing and engagement activities (asynchronous to new arrivals)

Request Intensity Growth:

```
At 10 users:    ~12 requests/sec (baseline)
At 110 users:   ~132 requests/sec
At 210 users:   ~252 requests/sec
At 310 users:   ~372 requests/sec
At 410 users:   ~492 requests/sec (starting to saturate)
At 510 users:   ~612 requests/sec (heavy degradation likely)
At 610 users:   ~732 requests/sec (critical threshold)
```

5.2.4 Performance Thresholds

Three performance boundaries define system behavior:

Threshold	Criteria	Load	Status
No Degradation	p95 < 500ms, error < 1%	~200 users (~240 req/sec)	✅ Healthy

Threshold	Criteria	Load	Status
With Degradation	p95 < 2000ms, error < 5%	200-450 users (~240-540 req/sec)	⚠ Degraded
Failure	p95 ≥ 2000ms OR error ≥ 5%	450+ users (540+ req/sec)	❌ Failed

No Degradation Zone (~200 users):

- All services operate with spare capacity
- PostgreSQL CPU < 40%, no lock contention
- Response times stable and predictable

Degradation Zone (200-450 users):

- PostgreSQL connection pool heavily utilized (80%+)
- Response times increase 3-4x (500ms → 1500-2000ms)
- Error rate 2-5%, primarily registration timeouts
- System remains functional but slower

Failure Zone (450+ users):

- PostgreSQL connection pool exhausted
- Cascading timeouts across services
- Error rate reaches 10-15%
- System unable to handle new requests

5.2.5 Expected Breaking Points

| Load | Users | Req/sec | p95 Response | Error Rate | System State | |------|-------|---------|--------------|------------|--------------|| | **Light** | 50 | 60 | 200ms | < 0.1% | Green - all systems nominal | | **Moderate** | 150 | 180 | 400ms | < 0.5% | Green - comfortable capacity | | **Heavy** | 250 | 300 | 1000ms | 1.5% | Yellow - degradation starting | | **Very Heavy** | 350 | 420 | 1500ms | 3% | Red - degradation significant | | **Critical** | 450 | 540 | 2000ms | 5% | Red - approaching failure | | **Failure** | 550+ | 660+ | 3000ms+ | 10%+ | Black - functionally broken |

5.2.6 Bottleneck Analysis

Primary: PostgreSQL Connection Pool

The shared PostgreSQL instance (User + Itinerary services) is the critical bottleneck. With only 20 connections per service (40 total), the queue saturates at 450+ users, causing 500-2000ms delays on simple queries (vs. 10-50ms baseline).

Mitigation: Implement **PgBouncer** connection pooling (200+ connections) and **read replicas** for Itinerary Service reads.

Secondary: MongoDB Indexing

MongoDB performs well at baseline but disk I/O becomes a constraint under 500+ users. Social Service queries benefit from indexed lookups on `(user_id, itinerary_id)`.

API Gateway Load:

Nginx remains stable up to 700+ req/sec, becoming CPU-bound above 1000 req/sec.

5.2.7 System Behavior Under Failure

The system exhibits **graceful degradation** rather than total failure:

- Casual browsing degrades first (least critical)
- Itinerary creation slows (medium priority)
- Existing reads remain responsive (API Gateway caching)
- Recovery occurs within 2-5 minutes once load decreases

Recommended Optimizations:

1. **Connection Pooling:** Deploy PgBouncer with 200+ connections
2. **Read-Write Splitting:** Route reads to replicas, writes to primary
3. **Cache Layer:** Add Redis for frequently accessed itineraries
4. **Circuit Breaker:** Reject new registrations when p95 > 1500ms
5. **Rate Limiting:** Queue excess registration requests

Appendices

[Any additional documentation, diagrams, or references to be added] **Description:** Continuous user growth simulation starting at 10 users and adding 20 users per minute continuously for 30+ minutes or until system failure. Growth rate: 20 users/min.

Results:

Metric	Value
Total Requests	244,939
Total Failures	95,022
Failure Rate	38.82%
Peak User Load	3,500 users
Response Times (Median)	8,400ms
Response Times (p95)	11,000ms
Response Times (p99)	17,000ms
Test Duration	~30 minutes (until time limit, not system failure)

Performance by User Load (Continuous Growth):

User Load	Failure Rate	p95 Response Time	Status

User Load	Failure Rate	p95 Response Time	Status
0-1,770 users	0.0-0.03%	9,000-10,000ms	✅ Healthy
1,800 users	0.03%	10,000ms	✅ Healthy
1,900 users	0.16-0.20%	10,000-11,000ms	⚠ Error Growth Begins
2,000 users	0.45-0.48%	11,000ms	⚠ Errors Accelerating
2,100 users	1.0%+	11,000ms	⚠ Sustained Error Growth
2,300 users	2.5-3%	12,000ms	⚠ Progressive Degradation
2,500 users	5.9%	12,000ms	⚠ Error Growth Continues
2,700 users	9.2%	12,000ms	⚠ Approaching Failure
2,800+ users	11-14%	12,000ms	❌ Significant Failures
3,000 users	14.84%	12,000ms	❌ Critical Load
3,500 users	38.75%	11,000ms	❌ Extreme Instability (spike)

Error Growth Pattern:

The system exhibits **continuous, linear error growth** from ~1,800 users onward rather than a discrete "degradation phase":

- **1,770 users:** Nearly 0% failures, responses stable at 9.5-10 seconds
- **1,800-1,900 users:** Error emergence point; failures begin rising (0.03% → 0.20%)
- **1,900-2,100 users:** Rapid acceleration; failures jump from 0.2% → 1.0%
- **2,100-3,000 users:** Steady growth at ~0.45-0.65% per 100 users added
- **3,000-3,500 users:** Escalation continues; failures grow from 14.84% → 38.75%

Key Observations:

1. **No Graceful Degradation:** Unlike Scenario B (which cycled), the lifetime test shows constant error growth. The system never reaches a "stable degraded state" but continuously worsens.
2. **Response Time Plateau:** p95 response times stabilize at 11,000-12,000ms after ~1,800 users and do not worsen further, indicating the bottleneck is connection exhaustion, not query processing time.
3. **Throughput Remains Stable:** RPS stays consistent at 180-300 throughout the load growth, confirming connections are limiting factor, not CPU/disk.

Critical Error Threshold:

Error emergence begins at ~**1,800 concurrent users** (0.03% failure rate, still negligible). Errors become significant at **2,000 users** (0.48%) and escalate rapidly. At **3,000 users**, failures reach 14.84%, and by **3,500 users**, spike to 38.75%.

Resource Utilization and Known Issues:

- **Database Connection Pool (Itinerary Service):** Prisma client connection pool exhaustion begins at ~1,800 users. Not all itinerary service pods could establish SQL connections due to pool timeout (100 connection limit per instance).

- **Error Message Observed:** "Timed out fetching a new connection from the connection pool. More info: Prisma Connection Pool Documentation (Current connection pool timeout: 10, connection limit: 100)"
- **Root Cause:** Itinerary service instances experiencing uneven connection pool initialization; some pods unable to acquire connections during high concurrency scenarios. This cascades into client request timeouts at API Gateway (10-second timeout).
- **Progressive Failure Mode:** As user load increases, more requests queue waiting for connections, resulting in linear error rate growth until system capacity exhaustion.

Analysis: System maintains healthy operation with excellent response times (9-10 second p95) up to ~1,770 concurrent users. Error emergence occurs at ~1,800 users due to connection pool exhaustion in Itinerary Service pods. Beyond this threshold, the system exhibits linear error growth: each additional 100 users adds approximately 0.45-0.65% additional failures. Response times plateau at 11-12 seconds, confirming the bottleneck is database connection management, not query execution. The test reaches 3,500 users before time limit (not system failure), at which point 38-39% of requests fail. The bottleneck is not query optimization but database connection pool management and uneven resource initialization across service instances. This is a classic connection pool saturation pattern requiring configuration adjustments (higher pool limits, connection reuse optimization, or pod replication) rather than query optimization.